Ex: 2.3.1

Find parametric equations for the line passing through $P(4,-1,2)$ and $Q=(6,-4,7)$

Using P for $\vec{r}_0 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$ $\quad \vec{d} = \overrightarrow{PQ} = \begin{bmatrix} 6-4 \\ -4-(-1) \\ 7-2 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$

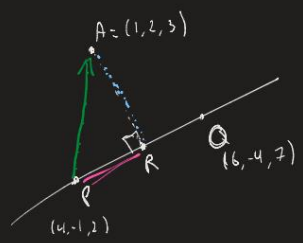
Vector Form: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$

$$x = 4 + 2t$$
$$y = -1 - 3t$$
$$z = 2 + 5t$$

Ex: 2.3.2

Find the distance from the point $A=(1,2,3)$ to the line in example 1 and also the coordinates of R.

$\overrightarrow{PQ} = [2,-3,5]$



$\overrightarrow{PA} = \begin{bmatrix} -3 \\ 3 \\ 1 \end{bmatrix}$

$\overrightarrow{PR} = \text{proj}_{\overrightarrow{PQ}} \overrightarrow{PA} = \left(\frac{\overrightarrow{PQ}\cdot\overrightarrow{PA}}{\overrightarrow{PQ}\cdot\overrightarrow{PQ}}\right)\overrightarrow{PQ} = \overbrace{\left(\frac{-6-9+5}{4+9+25}\right)}^{-\frac{5}{19}}[2,-3,5] = \left[-\frac{10}{19}, \frac{15}{19}, -\frac{25}{19}\right]$

$\overrightarrow{RA} = \overrightarrow{PA} - \overrightarrow{PR} = [-3,3,1] - \left[-\frac{10}{19}, \frac{15}{19}, -\frac{25}{19}\right] = \left[-\frac{47}{19}, \frac{42}{19}, \frac{44}{19}\right]$

Distance $= \|\overrightarrow{RA}\| = \sqrt{\left(-\frac{47}{19}\right)^2 + \left(\frac{42}{19}\right)^2 + \left(\frac{44}{19}\right)^2} \approx 4.05$

$\overrightarrow{PR} = \text{"R"} - \text{"P"}$

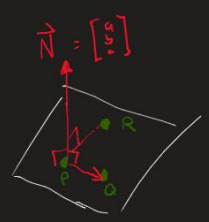
$= \left[-\frac{10}{19}, \frac{15}{19}, -\frac{25}{19}\right] = [x_R, y_R, z_R] - [4,-1,2]$

$$\begin{cases} -\frac{10}{19} = x_R - 4 & \to x_R = \frac{66}{19} \\ \frac{15}{19} = y_R + 1 & \to y_R = -\frac{4}{19} \\ -\frac{25}{19} = z_R - 2 & \to z_R = \frac{13}{19} \end{cases}$$

$R = \left(\frac{66}{19}, -\frac{4}{19}, \frac{13}{19}\right)$

Ex 2.3.3

Find a general form for the plane passing through the points $P = (0, 7, 1)$, $Q = (2, 2, 1)$ and $R = (4, -5, 0)$



- Find $\vec{PQ}$ and $\vec{PR}$

$$\vec{PQ} = \begin{bmatrix} 2 \\ -5 \\ 0 \end{bmatrix} \qquad \vec{PR} = \begin{bmatrix} 4 \\ -12 \\ -1 \end{bmatrix}$$

- Use $\vec{PQ} \times \vec{PR}$ to construct $\vec{N} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$$\vec{PQ} \times \vec{PR} = \begin{bmatrix} 2 \\ -5 \\ 0 \end{bmatrix} \times \begin{bmatrix} 4 \\ -12 \\ -1 \end{bmatrix} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -5 & 0 \\ 4 & -12 & -1 \end{vmatrix} = [5, 2, -4]$$

- Find d by using a point on the plane

$$5x + 2y - 4c = d$$

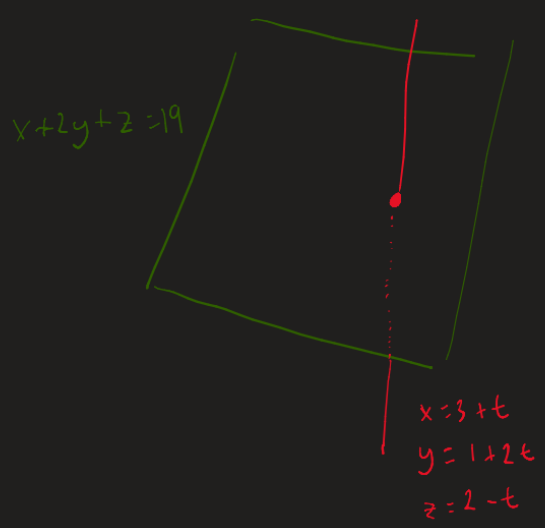
Point $P = (0, 7, 1)$: $5(0) + 2(7) - 4(1) = d$

$$d = 10$$

$$\boxed{5x + 2y - 4c = 10}$$

Ex 2.3.4

Find the point where the line $x = 3 + t$, $y = 1 + 2t$, $z = 2 - t$ intersects the plane $x + 2y + z = 19$



$$x + 2y + z = 19$$

$$(3+t) + 2(1+2t) + (2-t) = 19$$

$$3 + t + 2 + 4t + 2 - t = 19$$

$$4t = 19 - 7$$

$$t = \frac{12}{4} = 3$$

$$x = 3 + 3 = 6$$

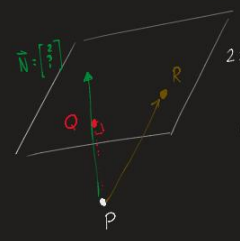
$$y = 1 + 2(3) = 7$$

$$z = 2 - 3 = -1$$

$$\boxed{\text{intersection point } (6, 7, -1)}$$

Ex 2.3.5

Find the distance between the plane $2x+3y+z=5$ and the point $P=(2,1,3)$. Find the point Q on the plane closest to P.



$2x+3y+z=5$

① Pick point R on the plane

pick $x=0$ $y=0$ $z=5$

$R = (0,0,5)$

② Form vector $\vec{PR}$ $\quad \vec{PR} = [-2,-1,2]$

$\vec{PR}\cdot\vec{N} = (-2)(2)+(-1)(3)+(2)(1) = -5$

$\vec{N}\cdot\vec{N} = 2^2+3^2+1^2 = 14$

③ $\vec{PQ} = \text{Proj}_{\vec{N}}\vec{PR} = \left(\frac{\vec{PR}\cdot\vec{N}}{\vec{N}\cdot\vec{N}}\right)\vec{N} = \frac{-5}{14}\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

$$= \begin{bmatrix} \frac{-10}{14} \\ \frac{-15}{14} \\ \frac{-5}{14} \end{bmatrix}$$

④ Distance from P to the plane $= \|\vec{PQ}\|$

$$\sqrt{\left(\frac{-10}{14}\right)^2+\left(\frac{-15}{14}\right)^2+\left(\frac{-5}{14}\right)^2} \approx 1.3363$$

⑤ $\vec{PQ} = \text{"}Q\text{"} - \text{"}P\text{"}$

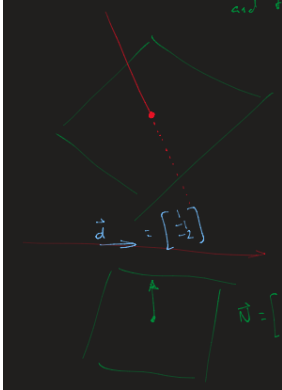
$$\Rightarrow \begin{bmatrix} \frac{-10}{14} \\ \frac{-15}{14} \\ \frac{-5}{14} \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$Q = \left(\frac{9}{7}, \frac{-1}{14}, \frac{37}{14}\right)$$

Example 2.3.6

Find the distance between the plane $x - 3y + 2z = 4$

and the line $\begin{cases} x = 4 + t \\ y = 3 - t \\ z = -2t \end{cases}$



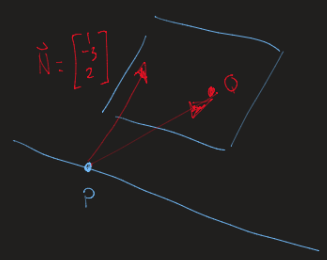
• Check whether line intersects

Method 1: $\vec{d} \cdot \vec{N} = 1 + 3 - 4 = 0 \Rightarrow$ orthogonal

$\Rightarrow$ line and plane parallel, hence no intersection

Method 2:

$$t + 2 - 3(3 - t) + 2(-2t) = 4$$
$$0t = 11$$
(No solution)



$\| \text{Proj}_{\vec{N}} \overrightarrow{PQ} \|$

$Q = (4, 0, 0)$ $\quad \vec{N} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$

$P = (2, 3, 0)$

$\overrightarrow{PQ} = [2, -3, 0]$

$$\text{Proj}_{\vec{N}} \overrightarrow{PQ} = \frac{\overrightarrow{PQ} \cdot \vec{N}}{\vec{N} \cdot \vec{N}} \vec{N} = \frac{11}{14} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{11}{14} \\ -\frac{33}{14} \\ \frac{22}{14} \end{bmatrix}$$

$$\overrightarrow{PQ} \cdot \vec{N} = 1(2) + (-3)(-3) + 0 = 11$$

$$\vec{N} \cdot \vec{N} = 1^2 + (-3)^2 + 2^2 = 14$$

$$\| \text{Proj}_{\vec{N}} \overrightarrow{PQ} \| = \sqrt{\left(\frac{11}{14}\right)^2 + \left(\frac{-33}{14}\right)^2 + \left(\frac{22}{14}\right)^2} \approx 2.9349$$

Ex 3.4.1

Gauss Jordan Elimination

Solve $\begin{cases} x - 4y = -2 \\ -3x + 2y = -4 \end{cases}$

$$\begin{array}{c} \left[\begin{array}{cc|c} 1 & -4 & -2 \\ -3 & 2 & -4 \end{array}\right] \\ \begin{array}{cc} \uparrow & \uparrow \\ x & y \end{array} \end{array} \xrightarrow{R_2 + 3(R_1)} \left[\begin{array}{cc|c} 1 & -4 & -2 \\ 0 & -10 & -10 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cc|c} 1 & -4 & -2 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{R_1 + 4R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

$x + 0_y = 2 \qquad x = 2$

$0_x + y = 1 \qquad y = 1$

Example 3.4.2

Solve $\begin{cases} 4x - 8y = 3 \\ -2x + 4y = -\frac{3}{2} \end{cases}$

$$\left[\begin{array}{cc|c} 4 & -8 & 3 \\ -2 & 4 & -\frac{3}{2} \end{array}\right] \xrightarrow{\left(\frac{1}{4}\right)R_1} \left[\begin{array}{cc|c} 1 & -2 & \frac{3}{4} \\ -2 & 4 & -\frac{3}{2} \end{array}\right]$$

$$\xrightarrow{R_2 + 2R_1} \left[\begin{array}{cc|c} 1 & -2 & \frac{3}{4} \\ 0 & 0 & 0 \end{array}\right] \text{ (RREF)}$$

leading variable: x

Free variable: $y = t$

$$x - 2\underbrace{\boxed{y}}_{t} = \frac{3}{4}$$

$$x = \frac{3}{4} + 2t$$

Ex. 3.4.3

$\begin{cases} x - y + 2z = 3 \\ x + 2y - z = -3 \\ 2y - 2z = 1 \end{cases}$

$ax + by + cz = d$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 1 & 2 & -1 & -3 \\ 0 & 2 & -2 & 1 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & -2 & 2 & 3 \\ 0 & 3 & -3 & -6 \\ 0 & 2 & -2 & 1 \end{array}\right] \xrightarrow{\left(\frac{1}{3}\right)R_2} \left[\begin{array}{ccc|c} 1 & -2 & 2 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 2 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & -2 & 2 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 5 \end{array}\right]$$

no solution

Ex 3.4.4

$$\begin{cases} 12x - 4y + z = -36 \\ 24x - 8y + 2z = -72 \\ -36x + 12y - 3z = 108 \end{cases}$$

Given $\left[\begin{array}{ccc|c} 12 & -4 & 1 & -36 \\ 24 & -8 & 2 & -72 \\ -36 & 12 & -3 & 108 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & \frac{1}{12} & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ infinite solutions

free variables: $y = s$
$z = t$

$$x - \frac{1}{3}s + \frac{1}{12}t = -3$$

$$x = \frac{1}{3}s - \frac{1}{12}t - 3$$

Ex 3.4.5

Find the coordinates of

$\vec{v} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$ with respect to (w, r, t)

$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 3 \\ -3 \\ -2 \end{bmatrix}$

$$k_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} + k_3 \begin{bmatrix} 3 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

$$\begin{cases} k_1 + 2k_2 + 3k_3 = -2 \\ k_2 - 3k_3 = 0 \\ k_1 + 4k_2 - 2k_3 = 3 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & -2 \\ 0 & 1 & -3 & 0 \\ 1 & 4 & -2 & 3 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$k_1 = 7$

$k_2 = -3$

$k_3 = -1$

Coordinate o $\vec{v}$ w.r.t $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are $(7, -3, -1)$

Example 3.5.1 Find the span of vectors $\vec{v}_1$ $\vec{v}_2$

(a) $\begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -4 \\ 13 \end{bmatrix}$ $\quad k_1\vec{v}_1 + k_2\vec{v}_2 = \vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\left[\begin{array}{cc|c} -1 & -4 & x \\ 3 & 13 & y \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 0 & -13x-4y \\ 0 & 1 & 3x+y \end{array}\right]$$

Always has solution

span $= \mathbb{R}^2$

(b) $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$$\left[\begin{array}{cc|c} 1 & 2 & x \\ 2 & 4 & y \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 2 & x \\ 0 & 0 & y-2x \end{array}\right]$$

has solution when $y-2x=0$

span$(\vec{v}_1, \vec{v}_2)$ = the line $y-2x=0$

(c) $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$

$$\left[\begin{array}{cc|c} 1 & 4 & x \\ 2 & 5 & y \\ 3 & 6 & z \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 0 & \frac{4}{3}y - \frac{5}{3}x \\ 0 & 1 & \frac{2}{3}x - \frac{1}{3}y \\ 0 & 0 & x - 2y + z \end{array}\right]$$

span$(\vec{v}_1, \vec{v}_2)$ = the plane $x-2y+z=0$

d) $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ Find span

General form of a plane

$ax + by + cz = d$

$$\left[\begin{array}{cc|c} 1 & 2 & x \\ 2 & 4 & y \\ 3 & 6 & z \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 2 & x \\ 0 & 0 & y-2x \\ 0 & 0 & z-3x \end{array}\right] \text{ planes}$$

solution only when $y-2x=0$ & $z-3x=0$

$$\boxed{\text{span}(\vec{v}_1, \vec{v}_2) = \text{line of intersection between two planes} \quad y-2x=0 \text{ and } z-3x=0}$$

Ex 3.5.2 For what value(s) of k (if any) is $\vec{u} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ in span $\left(\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ k \end{bmatrix} \right)$?

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 2 & k & 5 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1+k \end{array}\right]$$

$$1 + k = 0$$

$$\boxed{k = -1}$$

Ex 3.5.3
Express one of the vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ $v_3 = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$ as a linear combination of the other vectors if they are linearly dependant

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 0 & 5 & 0 \\ 2 & 2 & 4 & 0 \end{array}\right] \xrightarrow{\text{rref}} \begin{array}{c} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \\ \begin{array}{ccc} k_1 & k_2 & k_3 \end{array} \end{array}$$

k_3 is free variable
k_1, k_2 are leading variables

$k_3 = t$

$k_1 + 0k_2 + 5k_3 = 0$
$k_1 = -5t$

$k_2 = 3t$

$\therefore (-5t)\vec{v}_1 + (3t)\vec{v}_2 + (t)\vec{v}_3 = \vec{0}$

$-5\vec{v}_1 + 3\vec{v}_2 + \vec{v}_3 = \vec{0}$

$\vec{v}_3 = 5\vec{v}_1 - 3\vec{v}_2$

Ex 4.6.1

For $A = \begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 1 & 4 \\ 2 & 4 & 1 \end{bmatrix}$

compute $A + 2B$

$$2B = 2\begin{bmatrix} 5 & 1 & 4 \\ 2 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 2 & 8 \\ 4 & 8 & 2 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 10 & 2 & 8 \\ 4 & 8 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3+10 & 1+2 & 2+8 \\ 3+4 & 0+8 & 1+2 \end{bmatrix} = \begin{bmatrix} 13 & 3 & 10 \\ 7 & 8 & 3 \end{bmatrix}$$

Ex 4.6.2 $\quad AB + 4C^2$

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 2 \\ 1 & 4 \\ 4 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$AB = \underset{2\times 3}{\begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 1 \end{bmatrix}} \cdot \underset{3\times 2}{\begin{bmatrix} 5 & 2 \\ 1 & 4 \\ 4 & 1 \end{bmatrix}} = \begin{bmatrix} 3(5)+1(1)+2(4) & 3(2)+1(4)+2(1) \\ 3(5)+0(1)+1(4) & 3(2)+0(4)+1(1) \end{bmatrix}$$

$$= \begin{bmatrix} 24 & 12 \\ 19 & 7 \end{bmatrix}$$

$$C^2 = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2(2)+3(1) & 2(3)+3(2) \\ 1(2)+2(1) & 1(3)+2(2) \end{bmatrix}$$

$$C^2 = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$$

$$4C^2 = 4\begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 28 & 48 \\ 16 & 28 \end{bmatrix}$$

$$\begin{bmatrix} 24 & 12 \\ 19 & 7 \end{bmatrix} + \begin{bmatrix} 28 & 48 \\ 16 & 28 \end{bmatrix} = \begin{bmatrix} 52 & 60 \\ 35 & 35 \end{bmatrix}$$

Ex 4.6.3 Find the matrix A so that

$$(2A + I_2)^T = \begin{bmatrix} 9 & 4 \\ 2 & 3 \end{bmatrix}$$

$$2A + I_2 = \begin{bmatrix} 9 & 2 \\ 4 & 3 \end{bmatrix}$$

$$2A = \begin{bmatrix} 9 & 2 \\ 4 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \frac{1}{2} \begin{bmatrix} 8 & 2 \\ 4 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 1 \end{bmatrix}$$

Ex 4.6.4 For $B = \begin{bmatrix} 3 & k^2 \\ -k & -3 \end{bmatrix}$

find the values of k if

a) B is symmetric

$$B = B^T \Rightarrow \begin{bmatrix} 3 & -k \\ k^2 & -3 \end{bmatrix} = \begin{bmatrix} 3 & k^2 \\ -k & -3 \end{bmatrix}$$

$$\Rightarrow k^2 = -k \quad \Rightarrow k^2 + k = 0$$

$$k(k+1) = 0$$

$$k = 0 \text{ or } k = -1$$

b) $B\vec{v} = 6\vec{v}$ for $\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 3 & k^2 \\ -k & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 6 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} k^2 + 9 \\ -3k - 3 \end{bmatrix} = \begin{bmatrix} 18 \\ 6 \end{bmatrix}$$

$k^2 = 9$

$k = \pm 3$

$-3k - 9 = 0$

$\boxed{k = 3}$

c) $B^2 = I_2$ (we say B is a square root of I_2)

$$\begin{bmatrix} 3 & k^2 \\ -k & -3 \end{bmatrix} \begin{bmatrix} 3 & k^2 \\ -k & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 9 - k^3 & 3k^2 - 3k^2 \\ -3k + 3k & -k^3 + 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$